Mayank Kumar

Kanpur, Uttar Pradesh
github.com/Mayankkvv

Email : mayankkumar2325@gmail.com
Mobile : +91-7897590023
linkedin.com/in/mayank-kumar-004b5b280

Education

- **Indian Institute of Information Technology, Tiruchirappalli** — Tamil Nadu, India
  Bachelor of Technology in Electronics & Communication Engineering — *Aug 2023 – May 2027*

Experience

- **CODTECH IT Solutions** — Remote
  Software Engineering Intern — *May 2026 – Jul 2026*
  - *AI Workflow Automation Platform* – MERN, Redis, BullMQ, Socket.IO, Groq API
  - Engineered asynchronous AI workflows with Redis/BullMQ, reducing response latency by 95% (10s+ → <200ms) with real-time Socket.IO updates.
  - Built a React Flow workflow builder with JWT auth, MongoDB/Express backend, webhooks, and 5 OAuth2 integrations.
  - Implemented secure token management and deployed React frontend + Node.js API/worker on Vercel and Render.

Projects

- **Sync Engine – Real-Time Collaborative Document Editor** — Live Demo | GitHub
  React, Node.js, Express, WebSocket, MongoDB, JWT
  - Built a real-time collaborative document editor from scratch, implementing a custom CRDT algorithm to resolve concurrent multi-user edits with deterministic convergence, backed by WebSocket-based live sync, presence, and event-sourced version history.
  - Designed secure authentication with JWT and bcrypt, enforcing owner/collaborator access control independently across REST and WebSocket layers, with rate-limited login and a secure token-based password reset flow.
  - Validated correctness with 16 automated tests (Jest, Supertest) covering CRDT convergence, API authorization, and concurrent WebSocket client scenarios; deployed live on Vercel and Render.

- **DPI Engine – Deep Packet Inspection System** — GitHub
  C++17, Multithreading, CMake, Network Protocol Parsing, TLS/HTTP/DNS Inspection, PCAP, Python
  - Classified encrypted traffic across 20+ apps with zero decryption via a 4-way parallel DPI pipeline, achieving 100% flow identification (TLS SNI/HTTP/DNS) in test captures.
  - Engineered a zero-dependency, thread-safe packet pipeline (Reader → Load Balancer → Fast Path) using consistent-hash queues, enabling cross-platform blocking by IP/app/domain across concurrent flows.
  - Parsed raw PCAP captures at the Ethernet/IP/TCP/UDP layer with a custom byte-order abstraction layer, ensuring reliable protocol decoding across Linux, macOS, and Windows

Technical Skills

- **Languages**: Python, C++, Java, JavaScript
- **Core Concepts**: Data Structures & Algorithms, Object-Oriented Programming (OOP), System Design, Computer Networks, Operating Systems, Database Management Systems (DBMS)
- **Frontend**: HTML, CSS, React.js
- **Backend**: Node.js, Express.js, REST APIs, Flask
- **Databases**: MySQL, MongoDB, SQL
- **Tools & Platforms**: Git, GitHub, Docker, Postman, AWS

Achievements & Coding Profiles

- **Codeforces**: Rated Specialist, actively solving competitive programming problems.
- **CodeChef**: 3-Star rated, with a rating of 1602.
- **Decode X Hackathon**: Finalist, March 2026.
- **DSA Practice**: Solved 900+ Data Structures & Algorithms problems across platforms.